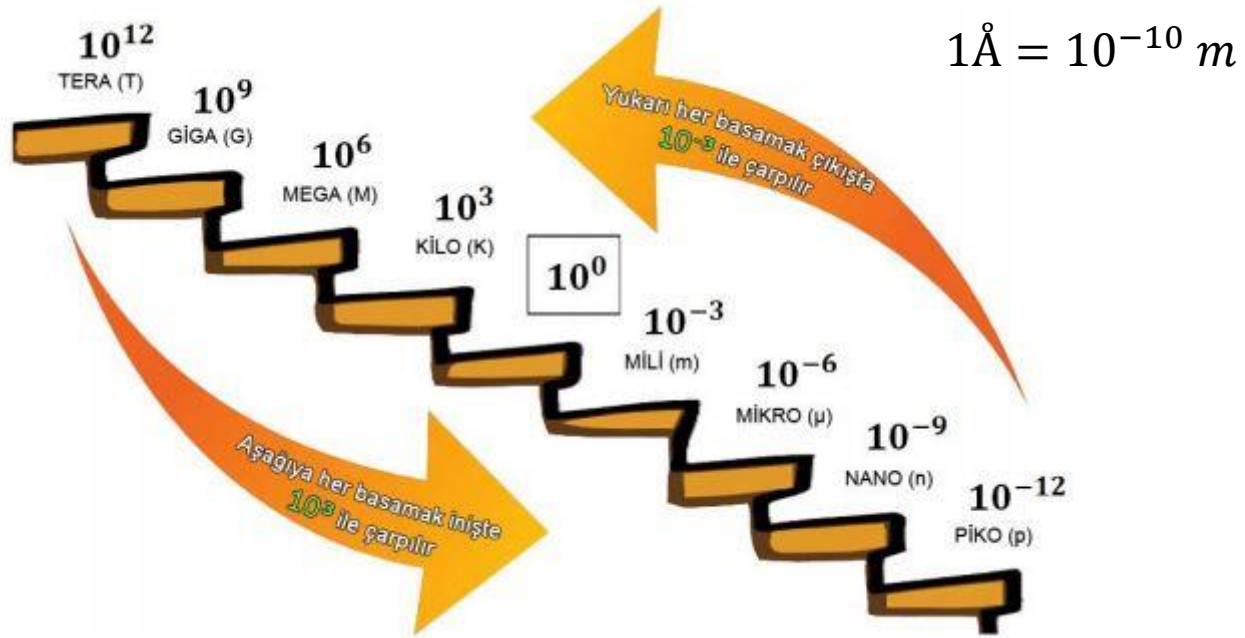


FİZ1951-MUHENDİSLER İÇİN YARIİLETKEN FİZİĞİ FORMÜL KAĞIDI



$$\epsilon_0 = 8.85 \times 10^{-12} \frac{F}{m}$$

$$h = 6.63 \times 10^{-34} J \cdot s$$

$$q = e = 1.6 \times 10^{-19} C$$

$$m_o = m_e = 9.11 \times 10^{-31} kg$$

$$k_B = 1.38 \times 10^{-23} J/K = 8.617 \times 10^{-5} eV/K$$

Elektron: $m_n = m_e$

Boşluk: Hol: Deşik: $m_h = m_b$

$$1 eV = 1.6 \times 10^{-19} J$$

$$n_i = CT^{3/2} e^{-E_g/2k_B T}$$

$$\sigma_i = n_i e (\mu_p + \mu_n)$$

$$C = 2 \left(\frac{2\pi k_B \sqrt{m_e^* m_h^*}}{h^2} \right)$$

$$n_i = p_i$$

$$n_i \cdot p_i = n_i^2 = p_i^2$$

$$1 \geq f_n(E, T) \geq 0$$

$$E_g = E_C - E_V$$

$$f_n(E, T) = \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}}$$

$$f_p(E, T) = \frac{1}{1 + e^{\frac{E_F - E}{k_B T}}}$$

$$N_{CB}(E) = \frac{1}{2\pi^2 \hbar^3} (2m_e^*)^{3/2} (E - E_C)^{1/2}$$

$$N_{VB}(E) = \frac{1}{2\pi^2 \hbar^3} (2m_h^*)^{3/2} (E_V - E)^{1/2}$$

$$n_i = \sqrt{N_C N_V} \exp\left(-\frac{E_g}{2k_B T}\right)$$

$$\frac{e}{k_B T} = \frac{\mu_p}{D_p} = \frac{\mu_n}{D_n}$$

$$N_C = 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{\frac{3}{2}}$$

$$N_V = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{\frac{3}{2}}$$

$$n_i = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right)$$

$$p_i = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

$$\vartheta_t = \sqrt{\frac{3k_B T}{m^*}}$$

$$E_{Fi} = E_V + \frac{1}{2} E_g + \frac{1}{2} k_B T \ln \frac{N_V}{N_C}$$

$$E_{Fi} = E_V + \frac{1}{2} E_g + \frac{3}{4} k_B T \ln \frac{m_h^*}{m_e^*}$$

$$\sigma = \frac{ne^2 \tau}{m_e^*}$$

$$\vec{x} = \vec{v}_s \Delta t$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqA\vartheta_s$$

$$J = \frac{I}{A}$$

$$\vec{J} = nq\vec{\vartheta}_s$$

$$V = IR$$

$$R = \rho \frac{l}{A}$$

$$\vec{v}_s = \mu \vec{E} = \frac{q\tau}{m} \vec{E}$$

$$\vec{J} = nq\mu \vec{E}$$

$$\sigma = nq\mu = ne\mu$$

$$q = e$$

$$\mu_n = \frac{e\tau_n}{m_n^*}$$

$$\mu_p = \frac{e\tau_p}{m_p^*}$$

$$\vec{J} = \sigma \vec{E} \quad \vec{E} = \frac{V}{l}$$

$$\rho = \frac{1}{ne\mu}$$

$$l = \mu_{\text{ort}} \tau$$

$$\mu_{\text{th}} = l / \tau$$

$$\sigma = 1/\rho \rightarrow S = 1/R$$

$$\vec{F}_e = q\vec{E} = m_o \vec{a}$$

$$\vec{F}_e = q\vec{E} + q\vec{E}_{Kr} = m^* \vec{a}$$

$$J_p = -eD_p \frac{dp}{dx} \quad J_n = eD_n \frac{dn}{dx}$$

$$D_n = \mu_n \frac{k_B T}{q} \quad D_p = \mu_p \frac{k_B T}{q}$$

$$\sigma_p = pq\mu_p \quad \sigma_n = nq\mu_n$$

$$J_{\text{Top}} = \left(en\mu_n \mathbf{E} + eD_n \frac{dn}{dx} \right) + \left(ep\mu_p \mathbf{E} - eD_p \frac{dp}{dx} \right)$$

$$N_A \text{ ve } N_D \gg n_i \quad \text{ve} \quad N_D > N_A \Rightarrow n \text{ tipi}$$

$$N_A \text{ ve } N_D \gg n_i \quad \text{ve} \quad N_A > N_D \Rightarrow p \text{ tipi}$$

$$n = n_i + N_D$$

$$N_D \gg n_i \rightarrow n \approx N_D$$

$$p = \frac{n_i^2}{N_D} \ll N_D$$

$$\sigma_n \approx N_D e \mu_n$$

$$p = p_i + N_A$$

$$N_A \gg p_i \rightarrow p \approx N_A$$

$$p = \frac{n_i^2}{N_A} \ll N_A$$

$$\sigma_p \approx N_A e \mu_p$$

$$E_{Fn} = E_C + k_B T \ln\left(\frac{n}{N_C}\right)$$

$$n = N_C \exp\left(-\frac{E_C - E_{Fn}}{k_B T}\right)$$

$$E_{Fp} = E_V - k_B T \ln\left(\frac{p}{N_V}\right)$$

$$p = N_V \exp\left(-\frac{E_{Fp} - E_V}{k_B T}\right)$$

$$n = \frac{(N_D - N_A)}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$\frac{1}{\mu_n^T} = \frac{1}{\mu_n^L} + \frac{1}{\mu_n^I}$$

$$\frac{qpV_H}{B_z} = \frac{I_x}{t}$$

$$E_{\text{Hall}} = v_s B, \quad v_s = \frac{I}{nqA}, \quad A = t \cdot W$$

$$V_{\text{Hall}} = R_{\text{Hall}} \frac{IB}{t}, \quad R_{\text{Hall}} = \frac{1}{nq}$$

$$q(\vec{\vartheta} \times \vec{B}) = q\vec{E}$$

$$qv_s B = qE_{\text{Hall}}$$